Every number except three is the path count of a best-connected pair

Vamshi Jandhyala

Abstract

For a graph G let $f(G)$ be the largest number of simple paths between two distinct vertices. No graph has $f(G) = 3$, a fact that a task at the 2020 International Olympiad in Informatics relies on. We show that every other positive integer is $f(G)$ for some graph. The graphs are four families in which a growing number of vertices are joined to the same two vertices x and y, giving $f = 2a$, $6a+1$, $6a+9$ and $6a+11$ for every $a \geq 1$, together with K_2, K_4, the wheel W_4 and one graph on six vertices. The families are analysed with a single observation: a simple path passes through x and y once each, so it can use at most one of the added vertices on the way. The construction is checked in the Lean proof assistant.

1 The question

Between two vertices of a tree there is exactly one simple path; around a cycle there are two. In the complete graph K_4 any two vertices are joined by 5 paths: the edge, two paths through one other vertex and two through both. So for the largest number $f(G)$ of simple paths between two distinct vertices of a graph G the values 1, 2 and 5 occur. Can you find a graph with $f(G) = 3$, or with $f(G) = 4$? Stop and try this before reading on.

The value 4 comes from the complete bipartite graph $K_{2,3}$: two vertices on the side of three have the two paths through one vertex of the other side and two more through all of the rest. The value 3 never occurs (Proposition 1). A task at the 2020 International Olympiad in Informatics [1] asks for a graph with prescribed numbers of paths, at most 3, between all pairs of vertices, and solving it uses this fact. In 2020 Mikhail Tikhomirov asked on MathOverflow [2] whether any other value is impossible, reporting a computer search in which every $k \leq 50$ other than 3 occurs on at most eight vertices.

Theorem 1. *For every positive integer $k \neq 3$ there is a graph G with $f(G) = k$.*

A comment on the question suggests gluing $K_{2,a}$ and $K_{2,b}$ at a vertex to obtain ab, which would leave only the primes. That does not quite work, because f is a maximum over *all* pairs and two vertices on the large side of $K_{2,a}$ already have $2a - 2$ paths between them. A corrected version is in Remark 3; the proof below does not need it.

2 A family where the count is a straight line

Start from a small graph H with two marked vertices x and y, and add a new vertices $m_1, \ldots, m_a$, each joined to x and y and to nothing else. Call the result G_a. Figure 1 shows one example.

Lemma 1. *Fix two vertices $u \neq v$ of G_a. The number of simple paths between u and v is $A + Bt$, where t is the number of new vertices other than u and v, and A, B depend only on H and on which vertices u and v are (a vertex of H, or a new vertex).*

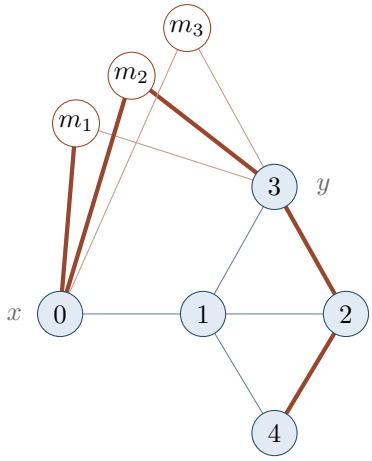


Figure 1: The $6a+1$ family with $a=3$: H is K_4 minus an edge on $\{1,2,3,4\}$ with the extra vertex 0, and the new vertices are joined to $x=0$ and $y=3$. The highlighted path from 4 to m_1 passes the new vertex m_2 as $3\,m_2\,0$; no path can pass two new vertices.

Proof. A new vertex inside a path has both of its path neighbours among its own neighbours, so the path runs $x\,m\,y$ or $y\,m\,x$ there. A simple path visits x once and y once, so at most one new vertex lies inside it. The paths with no new vertex inside are paths in H together with u and v; their number A does not depend on a. The paths with exactly one new vertex m inside correspond, for each choice of m, to the same set of paths with m renamed; so there are B of them for each of the t choices. □

Read as a statement about a: every pair of vertices of G_a has a path count that is a straight line in a (with $t=a$, $a-1$ or $a-2$ according as u and v include zero, one or two new vertices), and two values determine the line. So $f(G_a)$ is the largest of finitely many lines, and a claim such as $f(G_a)=6a+1$ for every $a\geq 1$ reduces to checking that one line is $6a+1$ and that every other line has slope at most 6 and lies below $6a+1$ at the first a where its pair exists. Table 1 lists four choices of H.

H (edges)	$x,\ y$	$f(G_a)$, every $a\geq 1$	a best pair
01, 12 (so $G_a=K_{2,a+1}$)	0, 2	$2a$	1, m_1
01, 12, 13, 14, 23, 24	0, 3	$6a+1$	4, m_1
01, 02, 03, 04, 05, 12, 23, 25, 34	2, 4	$6a+9$	1, m_1
01, 03, 04, 05, 12, 15, 23, 25, 34	0, 2	$6a+11$	1, 4

Table 1: Four families. The counts were found and checked by computer through Lemma 1; one is worked by hand below.

One line by hand. In the $6a+1$ family (Figure 1) take $u=4$ and a new vertex $m=m_1$. A path from 4 to m ends with the step $0\,m$ or $3\,m$.

- Ending $0\,m$: the path reaches 0 either from 1, along one of the 3 paths $4\,1$, $4\,2\,1$, $4\,2\,3\,1$, or from another new vertex m' that it reached from 3, along one of the 4 paths from 4 to 3 inside $\{1,2,3,4\}$. That gives $3+4(a-1)$.
- Ending $3\,m$: the path reaches 3 either inside $\{1,2,3,4\}$, in 4 ways, or as $1\,0\,m'\,3$ after one of the 2 paths from 4 to 1 that avoid 3. That gives $4+2(a-1)$.

The total is $6a+1$. Lemma 1 and the finite check show that no other pair does better.

Proof of Theorem 1. The family $2a$ gives every even number. Every odd number from 15 on is $6a+1$, $6a+9$ or $6a+11$ with $a \geq 1$, according to its residue modulo 6, and the $6a+1$ family also gives 7 and 13. That leaves 1, 5, 9 and 11: $f(K_2) = 1$, $f(K_4) = 5$, $f(W_4) = 9$ for the wheel with four rim vertices (two opposite rim vertices), and $f = 11$ for the graph with edges 01, 02, 05, 12, 13, 14, 15, 34, 45 (between 2 and 3). □

3 Why three is impossible

Proposition 1. *No graph has $f(G) = 3$.*

Proof. A path between u and v passes through the same blocks (maximal 2-connected pieces and bridges) in the same order, entering and leaving each at fixed cut vertices, so the number of paths between u and v is the product of the numbers of paths inside those blocks. A bridge contributes 1 and a cycle 2. If some pair has exactly 3 paths, then 3 is prime, so one block B has a pair with 3 paths; B is 2-connected and not a cycle. Such a graph contains a theta graph: three internally disjoint paths between two vertices. In a graph without multiple edges at most one of the three is a single edge, so two of them have interior vertices s and t. A path from s leaves along its own branch towards one of the two ends, then reaches t either directly along t's branch or round the third branch and back; that is $2 \times 2 = 4$ paths inside the theta alone. So $f(G) \geq 4$. □

Remark. Gluing two graphs at a vertex that is an end of a best pair in each multiplies f: a path between the two sides crosses the shared vertex once, and no pair on one side gains paths. With this, any $k = k_1 k_2$ is realised once k_1 and k_2 are. A computer search over all connected H with at most seven vertices found no family of slope 2 with odd values, and none of slope 4 with values 1 mod 4; that is why the odd families have slope 6.

Verification. Lemma 1 and the four families are checked in Lean 4 with Mathlib [3], with paths as lists of distinct vertices. The proof follows the text: a generator of all simple paths is proved exact, paths transfer between G_b and G_a along embeddings, a path has at most one interior new vertex, and the count in G_a is $N_b + (a-b)T$ for an embedded copy of a small G_b. The finitely many lines and the small graphs are then checked by evaluation. The final theorem states Theorem 1 for simple graphs, using only the standard axioms. Proposition 1 is proved here by hand and not formalised. A plain Python program, independent of the Lean code, recomputes every line. Everything is at `https://github.com/jvvk/mathematics/tree/main/every-path-count-but-three`.

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References

[1] International Olympiad in Informatics 2020, Day 1, task "Connecting Supertrees" (`supertrees`), `https://ioinformatics.org/files/ioi2020problem2.pdf`, accessed 11 October 2026.

[2] M. Tikhomirov, Largest number of simple paths between two vertices, MathOverflow question 371851 (16 September 2020), `https://mathoverflow.net/q/371851`, accessed 11 October 2026.

[3] The mathlib Community, The Lean mathematical library, in *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020)*, 367–381, doi:10.1145/3372885.3373824.

Vamshi Jandhyala, London, UK.